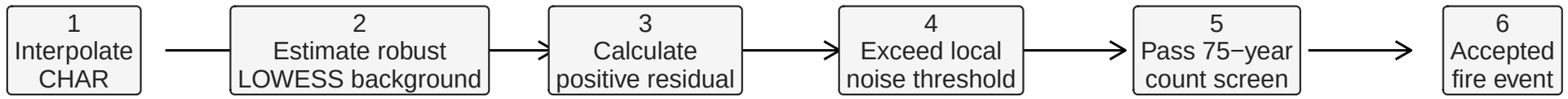
How an individual charcoal peak becomes an accepted fire event

The detection settings differ among cores; the later 1,000−year high−fire classification is a separate step.



	Engine	Chronology	Interpolation (years)	Background (years)	Noise percentile	Candidates	Count−screen rejected	Accepted
Sunrise Lake	tapas	published core chronology	48	800	95%	26	9	17
Shadow Lake	tapas	published core chronology	47	1000	95%	17	3	14
Little Sunrise Lake	tapas	published core chronology	42	500	95%	20	7	13
Panther Potholes	tapas	active PaleoACES Bacon median chronology	12	500	99%	66	0	66

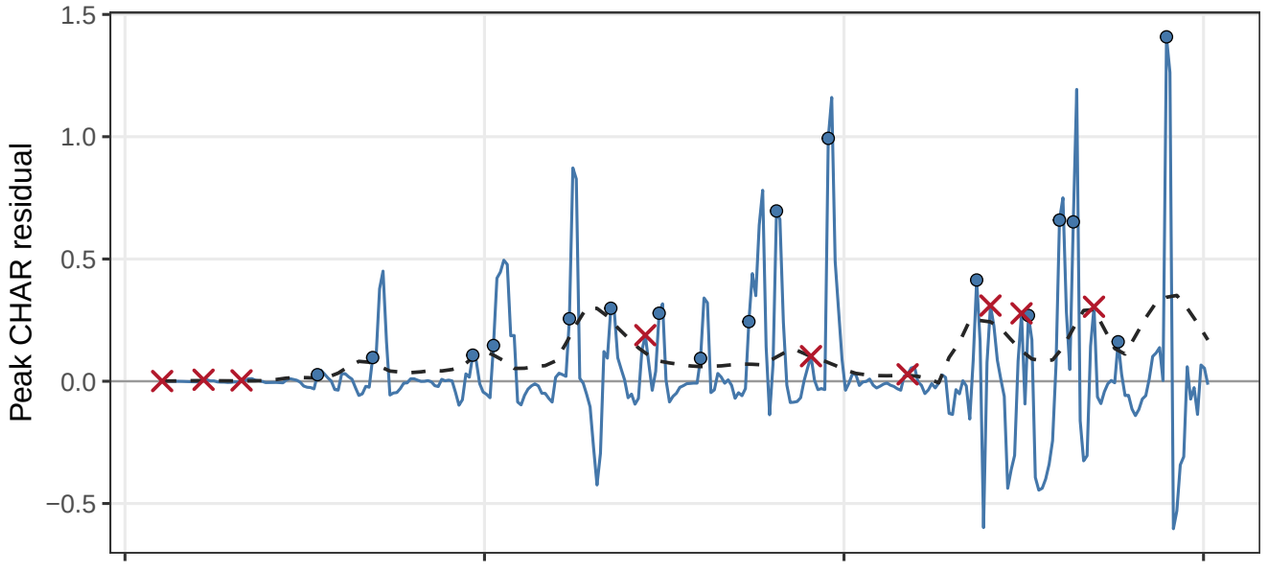
Accepted event = one distinct positive residual excursion above the local noise threshold that survives the count screen.
All four primary catalogs use tapas; each record retains its publication−specific temporal and threshold settings.
Accepted events are inferred charcoal episodes, not exact known ignition dates and not yet high−fire periods.

Fire−event detection across the four lake records

Dashed = threshold; filled circle = accepted; red X = rejected by count screen

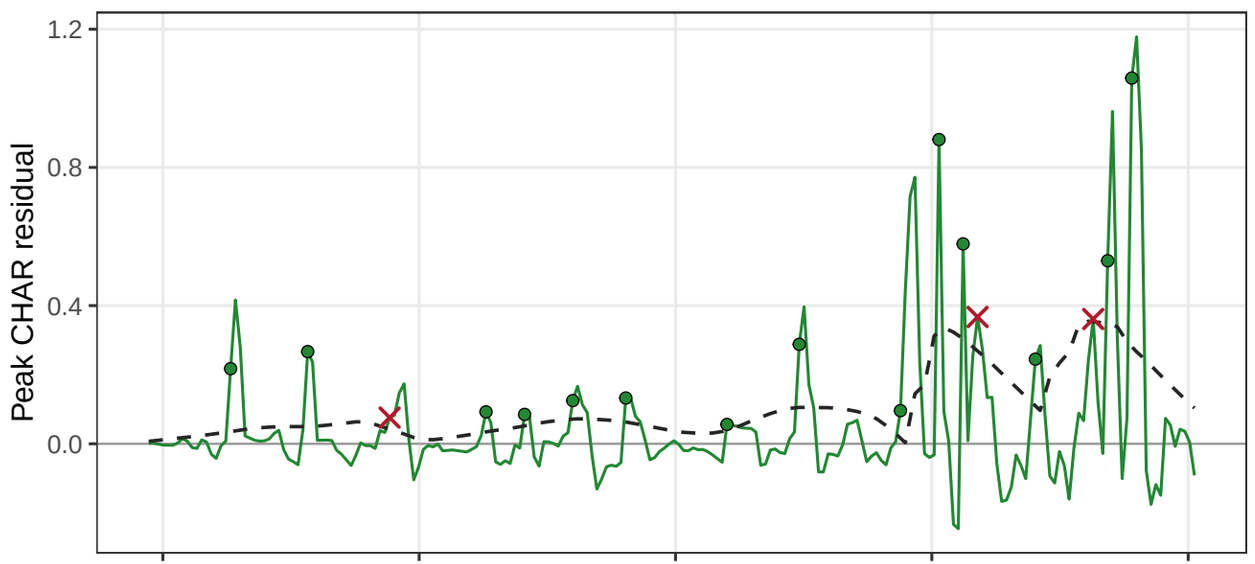
Sunrise Lake: local 95% noise threshold

Interpolation 48 yr | background 800 yr
17 accepted events | 9 rejected peaks



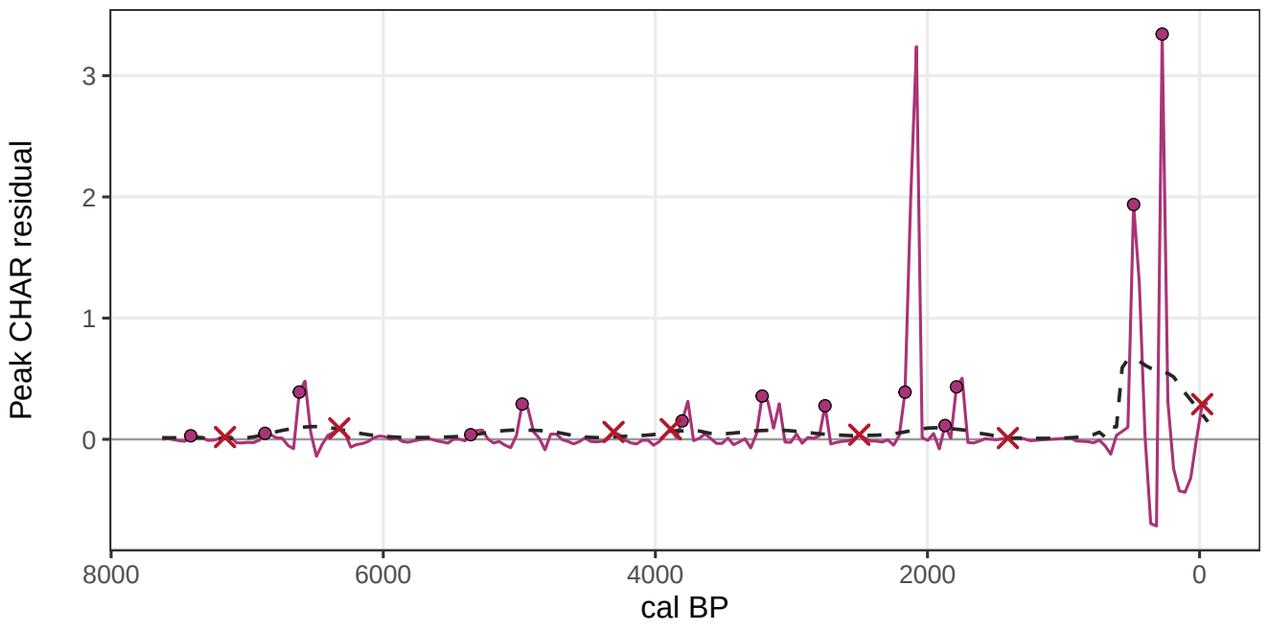
Shadow Lake: local 95% noise threshold

Interpolation 47 yr | background 1000 yr
14 accepted events | 3 rejected peaks



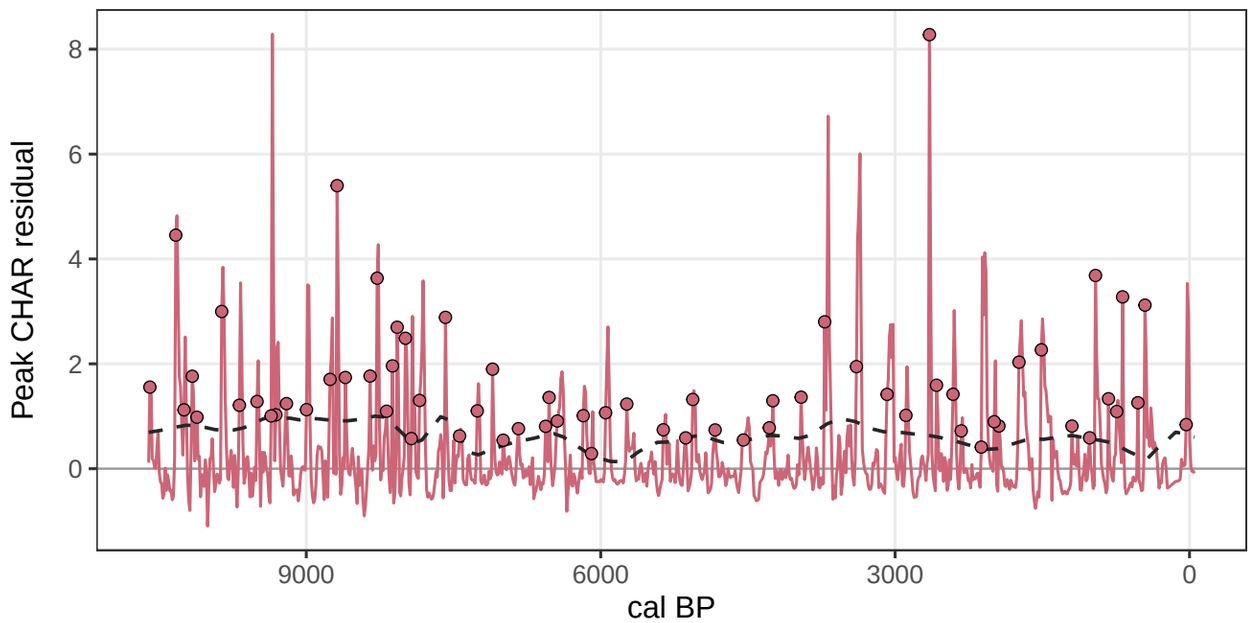
Little Sunrise Lake: local 95% noise threshold

Interpolation 42 yr | background 500 yr
13 accepted events | 7 rejected peaks

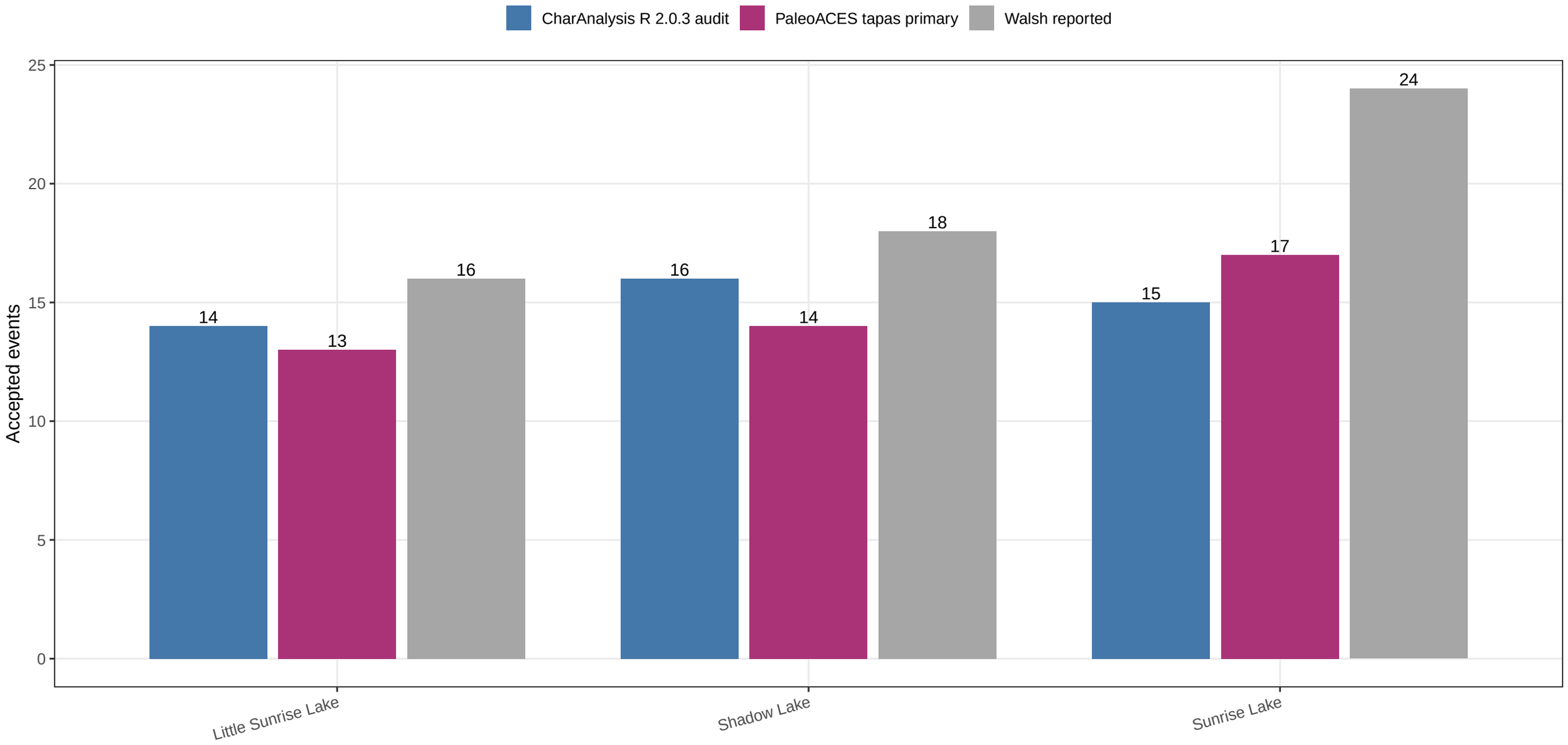


Panther Potholes: local 99% noise threshold

Interpolation 12 yr | background 500 yr
66 accepted events | 0 rejected peaks



Walsh publication totals, official audit, and harmonized primary catalog



The magenta tapas catalog is authoritative downstream. The official CharAnalysis run remains a reproducibility audit; neither implementation is tuned to force the published totals.